PS#7: Work and pressure - Answer key

1. (AC 6.4 Exercise 7) A cylindrical tank, buried on its side, has radius 3 feet and length 10 feet. It is filled completely with water whose weight density is 62.4 lbs/ft^3, and the top of the tank is two feet underground.

   (a) Set up, but do not evaluate, an integral expression that represents the amount of work required to empty the top half of the water in the tank to a truck whose tank lies 4.5 feet above ground.

$$\begin{aligned}\Delta W(h) &= \Delta F(h) \cdot D(h) \\ &= [\rho \cdot \Delta V(h)] \cdot D(h) \\ &= \left[\rho \cdot \left(10 \cdot 2\sqrt{9-h^2} \cdot \Delta h\right)\right] \cdot (9.5-h) \\ W_{\text{total}} &= \int_{h=0}^{3} \left[62.4 \cdot \left(10 \cdot 2\sqrt{9-h^2}\right)\right] \cdot (9.5-h)\, dh.\end{aligned}$$

   (b) With the tank now only half-full, set up, but do not evaluate an integral expression that represents the total force due to hydrostatic pressure against one end of the tank.

$$\begin{aligned}\Delta F &= P(h) \cdot \Delta A(h) \\ &= (62.4 \cdot h) \cdot (w(h) \cdot \Delta h) \\ &= (62.4 \cdot h) \cdot (2\sqrt{9-h^2} \cdot \Delta h) \\ F_{\text{total}} &= \int_{h=0}^{3} 62.4 \cdot h \cdot \left(2\sqrt{9-h^2}\right)\, dh.\end{aligned}$$

2. (Example 6.4.6) Trapezoid dam, three setups

Setup	Width of slice	Depth of slice	Bounds of integration
h down from water level	$w(h) = 84 - 6/5h$	$d(h) = h$	$0 < h < 20$
x down from top	$w(x) = 90 - 6/5x$	$d(x) = x - 5$	$5 < x < 25$
q up from bottom	$w(q) = 60 + 6/5q$	$d(q) = 20 - q$	$0 < q < 20$